Solve with us 2.1 - Not Graded

This assignment will not be graded and is only for practice.

1. Key Points:

Calculating minors and cofactors of a matrix:

- If AA is a square matrix, then the minor of the entry in the ii-th row and jj-th column (denoted by M_{ij} Mij) is the determinant of the submatrix formed by deleting the ii-th row and jj-th column.
- The ijij-th cofactor (denoted by C_{ij} Cij) is defined to be $(-1)^{i+j} M_{ij}$ (−1)i+jMij.

Calculating the determinant:

- $det(A) = \sum_{i=1}^{n} a_{ij} (-1)^{i+j} M_{ij} = \sum_{i=1}^{n} a_{ij} C_{ij}$ **det(A) = ∑i=1n aij(−1)i+jMij = ∑i=1n aijCij**

Let us consider a $3 \times 3$3 × 3 matrix AA as follows:

$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$A = [a11a21a31a12a22a32a13a23a33]

Calculating M_{11} M11 and C_{11} C11:

• Find out the sub-matrix by deleting the first row and the first column:

$\begin{bmatrix} \cdot & \cdot & \cdot \\ \cdot & a_{22} & a_{23} \\ \cdot & a_{32} & a_{33} \end{bmatrix}$ [⋅⋅⋅⋅a22a32⋅a23a33]

• Calculating the determinant of the sub-matrix:

$det(\begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix}) = a_{22} a_{33} - a_{32} a_{23}$ det([a22a32a23a33]) = a22a33 − a32a23

- $M_{11} = a_{22} a_{33} - a_{32} a_{23}$ M11 = a22a33 − a32a23
- $C_{11} = (-1)^{1+1} (a_{22} a_{33} - a_{32} a_{23}) = a_{22} a_{33} - a_{32} a_{23}$ C11 = (−1)1+1(a22a33 − a32a23) = a22a33 − a32a23

Calculating M_{23} M23 and C_{23}: C23 :

• Find out the sub-matrix by deleting the second row and the third column:

$\begin{bmatrix} a_{11} & a_{12} & \cdot \\ a_{31} & a_{32} & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix}$ [a11⋅a31a12⋅a32⋯]

• Calculating the determinant of the sub-matrix:

$det(\begin{bmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{bmatrix}) = a_{11} a_{32} - a_{31} a_{12}$ det([a11a31a12a32]) = a11a32 − a31a12

- $M_{23} = a_{11} a_{32} - a_{31} a_{12}$ M23 = a11a32 − a31a12
- $C_{23} = (-1)^{2+3} (a_{11} a_{32} - a_{31} a_{12}) = -a_{11} a_{32} + a_{31} a_{12}$ C23 = (−1)2+3(a11a32 − a31a12) = −a11a32 + a31a12

Calculating the determinant of A:

- Expanding with respect to the first row:

  $det(A) = a_{11}(-1)^{1+1}M_{11} + a_{12}(-1)^{1+2}M_{12} + a_{13}(-1)^{1+3}M_{13}$ det(A) = a11(−1)1+1M11 + a12(−1)1+2M12 + a13(−1)1+3M13

- Expanding with respect to the second row:

  $det(A) = a_{21}(-1)^{2+1}M_{21} + a_{22}(-1)^{2+2}M_{22} + a_{23}(-1)^{2+3}M_{23}$ det(A) = a21(−1)2+1M21 + a22(−1)2+2M22 + a23(−1)2+3M23

- Expanding with respect to the third row:

  $det(A) = a_{31}(-1)^{3+1}M_{31} + a_{32}(-1)^{3+2}M_{32} + a_{33}(-1)^{3+3}M_{33}$ det(A) = a31(−1)3+1M31 + a32(−1)3+2M32 + a33(−1)3+3M33

  The determinant can also be calculated by expanding along columns.

- Expanding with respect to the first column:
  $det(A) = a_{11}(-1)^{1+1}M_{11} + a_{21}(-1)^{2+1}M_{21} + a_{31}(-1)^{3+1}M_{31}$ det(A) = a11(−1)1+1M11 + a21(−1)2+1M21 + a31(−1)3+1M31

Consider a square matrix $A = \begin{bmatrix} -1 & 2 & 0 \\ 0 & -1 & 1 \end{bmatrix}$A = $\begin{bmatrix} & & \\ & & \\ -1 & 2 & -1 \end{bmatrix}$. Find out M_{23} M23.

[Hint: Find out the sub-matrix by deleting the second row and the third column: $\begin{bmatrix} -1 & 2 & \cdot \\ 0 & -1 & \cdot \end{bmatrix}$ $\begin{bmatrix} & & \\ -1 & 2 & \cdots \end{bmatrix}$

Calculate the determinant of the sub-matrix.]

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point

Consider a square matrix $A = \begin{bmatrix} -1 & 2 & 0 \\ 0 & -1 & 1 \end{bmatrix}$A = $\begin{bmatrix} & & \\ & & \\ -1 & 2 & -1 \end{bmatrix}$. Find out C_{32} C32.

[Hint: Find out the sub-matrix by deleting the third row and the second column: $\begin{bmatrix} -1 & 0 & \cdot \\ 2 & -1 & \cdot \end{bmatrix}$ $\begin{bmatrix} & & \\ & & \\ -1 & 2 & \cdots \end{bmatrix}$

Calculate the determinant of the sub-matrix to find M_{32} M32. To find the cofactor C_{23} C23 the sign has to be taken care of as follows: $C_{32} = (-1)^{3+2}M_{32}$ C32 = (−1)3+2M32]

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) -1

1 point

Consider a square matrix $A = \begin{bmatrix} -1 & 2 & 0 \\ 0 & -1 & 1 \end{bmatrix}$A = $\begin{bmatrix} -1 & 2 & 0 \\ 0 & -1 & 1 \end{bmatrix}$. Find out the determinant of AA.

[Hint: Expanding with respect to the first row: $det(A) = a_{11}(-1)^{1+1}M_{11} + a_{12}(-1)^{1+2}M_{12} + a_{13}(-1)^{1+3}M_{13}$ det(A) = a11(−1)1+1M11 + a12(−1)1+2M12 + a13(−1)1+3M13

Expanding with respect to any row will also give you the answer]

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -2

1 point

1 point

Consider the following square matrices:

$A = \begin{bmatrix} 2013 & 2014 & 2015 \\ 2016 & 2017 & 2022 \end{bmatrix}$, $B = \begin{bmatrix} 2016 & 2017 & 2022 \\ 2013 & 2014 & 2015 \end{bmatrix}$ and $C = \begin{bmatrix} 4032 & 4034 & 4044 \\ 2013 & 2014 & 2015 \end{bmatrix}$A = $\begin{bmatrix} 2013 & 2014 & 2015 \\ 2016 & 2017 & 2022 \\ 2019 & 2020 & 2021 \end{bmatrix}$, B = $\begin{bmatrix} 2016 & 2017 & 2022 \\ 2013 & 2014 & 2015 \\ 2019 & 2020 & 2021 \end{bmatrix}$ and C = $\begin{bmatrix} 4032 & 4034 & 4044 \\ 2013 & 2014 & 2015 \\ 2019 & 2020 & 2021 \end{bmatrix}$

Choose the set of correct options.

Hint:

- BB can be obtained by interchanging the first two rows.
- CC can be obtained by multiplying 2 to the first row of BB.

☐ $det(B) = det(A)$det(B) = det(A).

☐ $det(A) = -det(B)$det(A) = −det(B).

☐ $det(C) \neq -2det(B)$det(C) = −2det(B).

☐ $det(C) = -2det(A)$det(C) = −2det(A)

No, the answer is incorrect.

Score: 0

Feedback:

• Sign of the determinant changes due to interchange of two rows.

• If a scalar cc is multiplied with a row of a matrix, then the determinant of the new matrix will be cc times the determinant of the earlier matrix.

Accepted Answers:

$det(A) = -det(B)$det(A) = −det(B).

$det(C) \neq -2det(B)$det(C) = −2det(B).
$det(C) = -2det(A)$det(C) = −2det(A)

2. Key Points:

Cramers' Rule: (This rule is applied to any system of linear equations with nn equations and nn variables. Here we recall the method for a system of linear equations with 33 equations and 33 variables.)

- Consider a system of linear equations as follows: $\begin{matrix} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3 \end{matrix}$a11x1 + a12x2 + a13

x3 = b1a21x1 + a22x2 + a23x3 = b2a31x1 + a32x2 + a33x3 = b3

- Let the matrix representation of the above system be $Ax = b$Ax = b, where $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$A = [a11

a21a31a12a22a32a13a23a33], $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$x = [x1x2x3], and $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$b = [b1b2b3].

- Let A_{x_i} Axi be the matrix obtained by replacing the ii-th column of AA ($i.e., \begin{bmatrix} a_{1i} \\ a_{2i} \\ a_{3i} \end{bmatrix}$)(i.e., [a1ia2ia3i]) by bb, for $i = 1, 2, 3$i = 1, 2, 3.

$A_{x_1} = \begin{bmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{bmatrix}$Ax1 = [b1b2b3a12a22a32a13a23a33]

$A_{x_2} = \begin{bmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{bmatrix}$Ax2 = [a11a21a31b1b2b3a13a23a33]

$A_{x_3} = \begin{bmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{bmatrix}$Ax3 = [a11a21a31a12a22a32b1b2b3]

If $det(A) \neq 0$det(A) = 0, then the solutions to the above system are $x_i = \frac{det\ A_{x_i}}{det\ A}$xi = det Adet Axi, for $i = 1, 2, 3$i = 1, 2, 3. i.e., $x_1 = \frac{det\ A_{x_1}}{det\ A}$x1 = det Adet Ax1, $x_2 = \frac{det\ A_{x_2}}{det\ A}$x2 = det Adet Ax2, $x_3 = \frac{det\ A_{x_3}}{det\ A}$x3 = det Adet Ax3

Consider a system of linear equations $\begin{matrix} x_1 + x_3 = 1 \\ -x_1 + x_2 - x_3 = 1 \\ -x_2 + x_3 = 1 \end{matrix}$x1 + x3 = 1−x1 + x2 − x3 = 1−x2 + x3 = 1

Let matrix representation of the above system be $Ax = b$Ax = b, where $A = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$A = $\begin{bmatrix} 1 & 0 & 1 \\ -1 & 0 & -1 \\ 0 & -1 & 1 \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$x = $\begin{bmatrix} x1 \\ x2 \\ x3 \end{bmatrix}$ and $b = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$b = $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. Let A_{x_i} Axi be the matrix obtained by replacing the ii-th column of AA (i.e., $\begin{bmatrix} a_{1i} \\ a_{2i} \\ a_{3i} \end{bmatrix}$ $\begin{bmatrix} a1i \\ a2i \\ a3i \end{bmatrix}$) by bb, for $i = 1, 2, 3$i = 1, 2, 3.

Use the above information to answer the following questions(5,6):

1 point

Choose the set of correct options

☐ $A_{x_1} = \begin{bmatrix} 1 & 1 & 0 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$Ax1 = $\begin{bmatrix} 1 & -1 & 0 \\ 1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$

☐ $A_{x_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$Ax1 = $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$

☐ $A_{x_2} = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$Ax2 = $\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$

☐ $A_{x_3} = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix}$Ax3 = $\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$A_{x_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$Ax1 = $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$

A_{x_3} = [1 0 1; −1 1 1; 0 −1 1]Ax3 = [1 −1 0; 0 1 −1; 1 1 1]

1 point

Choose the set of correct options about the solutions to this system.

- [ ] $x_1 = -2$x1 = −2.
- [ ] $x_2 = -2$x2 = −2.
- [ ] x_3 = 3x3 = 3.
- [ ] None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$x_1 = -2$x1 = −2.

x_3 = 3x3 = 3.

3. Key Points:

Calculating the inverse of a matrix A:

- Find out the cofactor matrix CC, whose ijij-th element is C_{ij} Cij: the ijij-th cofactor of AA.
- Adjoint of AA is transpose of CC.
- Calculate the determinant of AA and check whether it is non-zero or not.
- If determinant is non-zero then the inverse of AA exists and is given by

$$A^{-1} = \frac{1}{det(A)} adj(A)$$A−1 = det(A)1adj(A).

1 point

Which of the following is the cofactor matrix of [3 2 3; 1 0 1; 2 1 1] [3 1 2; 2 0 1; 2 1 1] ?

[Hint: Recall the steps to calculate the cofactors, given in Key Points 1.]

- ○ [1 1 1; −2 0 2] [1 −1 −2; −1 −3 0; 1 −1 2]
- ○ [−1 1 2; 1 −3 0; 1 1 −2] [−1 1 1; 1 −3 1; 2 0 −2]
- ○

$\begin{bmatrix} -1 & 1 & 1 \\ 1 & -3 & 1 \\ 2 & 0 & -2 \end{bmatrix}\begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix}_{-1121-3011-2}$

○

$\begin{bmatrix} 1 & -1 & -1 \\ -1 & 3 & -1 \\ -2 & 0 & 2 \end{bmatrix}\begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix}_{1-1-2-130-1-12}$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\begin{bmatrix} -1 & 1 & 1 \\ 1 & -3 & 1 \\ 2 & 0 & -2 \end{bmatrix}\begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix}_{-1121-3011-2}$

1 point

Which of the following is the adjoint matrix of $\begin{bmatrix} 3 & 2 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}\begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix}_{312201311}$?

[Hint: Find the transpose to the cofactor matrix.]

○

$\begin{bmatrix} 1 & -1 & -2 \\ 1 & -3 & 0 \\ 1 & -1 & 2 \end{bmatrix}\begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix}_{111-1-3-1-202}$

○

$\begin{bmatrix} -1 & 1 & 1 \\ 1 & -3 & 1 \\ 2 & 0 & -2 \end{bmatrix}\begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix}_{-1121-3011-2}$

○

$\begin{bmatrix} -1 & 1 & 2 \\ 1 & -3 & 0 \\ 1 & 1 & -2 \end{bmatrix}\begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix}_{-1111-3120-2}$

○

$\begin{bmatrix} 1 & -1 & -2 \\ -1 & 3 & 0 \\ -1 & -1 & 2 \end{bmatrix}\begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix}_{1-1-1-13-1-202}$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\begin{bmatrix} -1 & 1 & 2 \\ 1 & -3 & 0 \\ 1 & 1 & -2 \end{bmatrix} \begin{bmatrix} -1 & 1 & 1 & 1 \\ -3 & 1 & 2 & 0 \\ -1 & 2 & 0 & -2 \end{bmatrix}$

Find out the determinant of $\begin{bmatrix} 3 & 2 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 & 2 & 2 \\ 0 & 1 & 3 & 1 \\ 1 & 2 & 0 & 1 \end{bmatrix}$.

[Hint: Recall the steps to calculate the determinant, given in Key Point 1.]

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point
1 point

Which of the following is the inverse matrix of $\begin{bmatrix} 3 & 2 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 & 2 & 2 \\ 0 & 1 & 3 & 1 \\ 1 & 2 & 0 & 1 \end{bmatrix}$?

[Hint: If determinant is non-zero then the inverse of AA exists and is given by $A^{-1} = \frac{1}{det(A)} adj(A)$ A−1 = det(A)1adj(A).]

○ $\frac{1}{2}\begin{bmatrix} 1 & -1 & -2 \\ 1 & -3 & 0 \\ 1 & -1 & 2 \end{bmatrix} \frac{1}{2}\begin{bmatrix} 1 & -1 & -1 & -2 \\ -3 & -1 & 0 & 2 \end{bmatrix}$

○ $\frac{1}{2}\begin{bmatrix} -1 & 1 & 1 \\ 1 & -3 & 1 \\ 2 & 0 & -2 \end{bmatrix} \frac{1}{2}\begin{bmatrix} -1 & 1 & 2 & 1 \\ -3 & 0 & 1 & -2 \end{bmatrix}$

○ $\frac{1}{2}\begin{bmatrix} -1 & 1 & 2 \\ 1 & -3 & 0 \\ 1 & 1 & -2 \end{bmatrix} \frac{1}{2}\begin{bmatrix} -1 & 1 & 1 & 1 \\ -3 & 1 & 2 & 0 \\ -1 & 2 & 0 & -2 \end{bmatrix}$

○ $\frac{1}{2}\begin{bmatrix} 1 & -1 & -2 \\ -1 & 3 & 0 \\ -1 & -1 & 2 \end{bmatrix} \frac{1}{2}\begin{bmatrix} 1 & -1 & -1 & -1 \\ 3 & -1 & -2 & 0 \end{bmatrix}$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\frac{1}{2}\begin{bmatrix} -1 & 1 & 2 \\ 1 & -3 & 0 \\ 1 & 1 & -2 \end{bmatrix}\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}\begin{bmatrix} -1 & 1 & 1 \\ 1 & -3 & 1 \\ 2 & 0 & -2 \end{bmatrix}$

1 point

Choose the set of correct options.

Hint:

- $A^{-1} = \frac{1}{det(A)} adj(A)$A−1 = det(A)1adj(A).
- $det(A^{-1}) = \frac{1}{det(A)}$det(A−1) = det(A)1

☐ If $A = A^{-1}$ A = A−1, then $det(A)$det(A) must be 1.

☐ If $A = A^{-1}$ A = A−1, then AA must be the identity matrix.

☐ If $A = A^{-1}$ A = A−1, then $A^2 = I$A2 = I.

☐